BIO PRO
Mold & IAQ Testing

Mold Assessment Report & Remediation Protocol

Client: Merritt Island Wildlife Refuge

Address: 1963 Refuge Headquarters Road,
PO Box 2683
Titusville, FL 32780

Inspected By: Jared Davis

Inspection Date: May 22, 2026



1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

Merritt Island Wildlife Refuge
1963 Refuge Headquarters Road,
PO Box 2683
Titusville, FL 32780

Inspection Date: May 22, 2026

Thank you for choosing Bio Pro to perform a mold assessment at your property. After conducting a thorough on-site investigation and collecting site-specific microbial samples, we have provided a detailed environmental analysis and recommendations for proper remediation per IICRC-S520 standards.

Sampling and analysis were conducted in accordance with applicable ASTM standards, where available, and interpreted using recognized industry guidelines (IICRC S520, EPA).

At Bio Pro, we're dedicated to resolving your indoor environmental issues and we strive to provide unrivaled customer service. Please feel free to contact us with any questions that you may have and we'd be happy to go over your report in detail over the phone.

Thank you for trusting Bio Pro with your home or business!

Jared Davis

Owner & Licensed Mold Assessor
Bio Pro: Mold and IAQ Testing
(321) 622-0830
Office@BioProFL.com

BIO PRO
Mold & IAQ Testing

1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

Understanding Your Report

Intro to Mold:

Mold is a natural part of the environment and plays an essential role in breaking down organic matter. However, not all molds are harmless, and certain conditions in your home, like excessive moisture or water damage, can foster the growth of molds that may impact your health and the integrity of your building.

Molds We Focus On:

At Bio Pro, our inspection and testing services are designed to identify mold species that are not typically part of a healthy indoor environment. These molds can indicate underlying issues in your home, such as water leaks or high humidity, and may post health risks. Here are the key mold types we specifically look for.

Aspergillus/ Penicillium:	This group of molds is common in damp or water damaged buildings. They can cause allergic reactions and respiratory issues. Regular presence of these molds can indicate humidity problems or an ongoing moisture issue.
Chaetomium:	Often found in water damaged buildings, Chaetomium species are known for their musty odor. They are associated with water damage and can degrade indoor air quality significantly.
Stachybotrys:	Commonly referred to as "black mold", Stachybotrys is particularly concerning because of its potential to produce toxins. This type of mold is usually found in buildings with serious water damage and requires immediate attention.
Cladosporium:	Frequently identified in air-conditioning systems and HVAC ducts, this mold can thrive in both cool and warm areas. Known for triggering allergic reactions and respiratory problems such as asthma attacks. Its presence often indicates a need for maintenance or cleaning of HVAC systems to prevent health issues and spread.

1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

Why These Molds Matter

The presence of these specific molds can signify more severe issues in your property, such as:

Moisture & Water Damage	Persistent moisture creates ideal conditions for harmful molds. Identifying and resolving the source of moisture is crucial to prevent further mold growth and structural damage
Health Risks	Exposure to toxic molds can lead to various health problems, especially for individuals with respiratory issues, allergies, or compromised immune systems.

What To Do Next?

If any of these molds are identified in your home, it's important to take steps to address the underlying causes of their growth. Our report includes detailed findings and recommendations for remediation to ensure your indoor environment is safe and healthy.

For further assistance or to discuss your mold report in detail, please contact us at (321) 622-0830.

1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

General Information

Bio Pro conducted a mold inspection on May 22, 2026 to assess potential mold growth and indoor air quality. The inspection included a visual evaluation and environmental sampling. Findings and recommendations follow applicable industry guidelines.

Client-Reported Concerns

The client is an office building for the Merritt Island Wildlife Refuge and has experienced several previous water damage issues.

Inspection Findings

• **ATTIC:** localized areas of visible mold growth were observed on the roof decking, confirmed by sampling to include Stachybotrys

• **AMBIENT AIR QUALITY:** air samples collected from multiple office areas were not elevated, indicating no widespread airborne mold issue at the time of inspection

• **HUMIDITY CONDITION**S: indoor relative humidity was measured at approximately 70%, which is above recommended levels and can promote mold growth

• **HVAC SYSTEM:** visible mold growth was present on several AC registers and on the interior lining of one flex duct, likely due to high humidity and condensation

• **WATER DAMAGE**: the north side of the building, particularly in the east and west offices, showed signs of severe water intrusion and damage, with visible mold growth on lower drywall sections

• **WALL CAVITIES:** sampling confirmed mold contamination within the wall cavities of affected areas, while adjacent shared walls tested negative, indicating localized contamination

4795 Enterprise Ave
Naples, Florida 34104
login@daanelabs.com
www.daanelabs.com

REPORT CODE: M-JADA-265637

Company	Bio Pro	Project Name	FWRH	
Address	1275 S Patrick Dr H4, Satellite Beach, FL 32937			
Contact	Joy Hoover	Project Address	FWRH - Refuge Headquarters, Merritt Island, FL 32955	
Phone	321-622-0830			
Email	office@bioprofl.com	Analyzed by/ Date	SR	05/26/26

Lab ID Number	M-JADA-265571-001	M-JADA-265637-001	M-JADA-265637-002	M-JADA-265637-003
Collection Date	05/22/26	05/22/26	05/22/26	05/22/26
Volume	75	75	75	75
Location	Outside	Conf. Table	East Office	West Office
RESULT†	**CONTROL**	**NOT ELEVATED**	**NOT ELEVATED**	**NOT ELEVATED**
% Slide Analyzed	100	100	100	100

Spore Identification	Raw Count	Spore/m³	% of Total	Raw Count	Spore/m³	% of Total	Raw Count	Spore/m³	% of Total	Raw Count	Spore/m³	% of Total
Aspergillus/ Penicillium		0	0	5	67	14		0	0	2	27	15
Chaetomium		0	0		0	0		0	0		0	0
Stachybotrys		0	0		0	0		0	0		0	0
Alternaria		0	0		0	0		0	0		0	0
Arthrinium		0	0		0	0		0	0		0	0
Ascospores	7	93	14	4	53	11	2	27	8	3	40	23
Basidiospores	30	400	59		0	0	4	53	15		0	0
Cladosporium	14	187	27	23	307	64	20	267	77	8	107	62
Cercospora		0	0		0	0		0	0		0	0
Curvularia		0	0		0	0		0	0		0	0
Dreschlera/ Bipolaris		0	0		0	0		0	0		0	0
Epicoccum		0	0		0	0		0	0		0	0
Fusarium		0	0		0	0		0	0		0	0
Ganoderma		0	0		0	0		0	0		0	0
Memnoniella		0	0		0	0		0	0		0	0
Myxomycetes/ Smut		0	0	4	53	11		0	0		0	0
Nigrospora		0	0		0	0		0	0		0	0
Pithomyces		0	0		0	0		0	0		0	0
Rust		0	0		0	0		0	0		0	0
Spegazzinia		0	0		0	0		0	0		0	0
Torula		0	0		0	0		0	0		0	0
Ulocladium		0	0		0	0		0	0		0	0
Other		0	0		0	0		0	0		0	0
Total Fungi	**51**	**680**	**100**	**36**	**480**	**100**	**26**	**347**	**100**	**13**	**173**	**100**
Hyphal Fragment		0	N/A		0	N/A		0	N/A		0	N/A
Background Debris (1-5)*		1			3			3			3	

Background Debris is a subjective assessment of the debris level (i.e., house dust) present in the sample, ranked from 1 to 5. A higher number corresponds to a higher level of debris.

*Higher Background Debris may interfere with the analyst's ability to identify spores

1 = 0-5% debris; 2 = 5-25% debris; 3 = 25-75% debris; 4 = 75-90% debris; 5 = 90-100% debris

AIR RESULT KEY†

ELEVATED	The concentration of spores in this sample exceeds the upper HHS threshold, which indicates that an indoor mold source is LIKELY.
POSSIBLE	The concentration of spores in this sample exceeds the lower HHS threshold, which indicates that an indoor mold source is POSSIBLE.
NOT ELEVATED	The concentration of spores in this sample does not exceed the lower HHS threshold, which indicates that an indoor mold source is UNLIKELY.
CONTROL	The indoor samples are compared to the control sample to determine whether there may be an indoor mold source.

SURFACE RESULT KEY†

GROWTH LIKELY	ASSOCIATED WITH IICRC S520 CONDITION 3: ACTIVE MOLD	**SURFACE LEVELS**	
GROWTH POSSIBLE	ASSOCIATED WITH IICRC S520 CONDITION 2: SETTLED SPORES	Rare: 1-9 spores	Low: 10-100 spores
GROWTH UNLIKELY	ASSOCIATED WITH IICRC S520 CONDITION 1: NORMAL ECOLOGY	Medium: 101-1,000 spores	High: >1,000 spores

The Laboratory is not responsible for project sampling. Customer provided information: Project Name, Project Number, Project ID, Project Address, Collection Date, Volume, and Location

Method : ASTM D7391

4795 Enterprise Ave
Naples, Florida 34104
login@daanelabs.com
www.daanelabs.com

REPORT CODE: M-JADA-265637

Company	Bio Pro					Project Name			FWRH	
Address	1275 S Patrick Dr H4, Satellite Beach, FL 32937									
Contact	Joy Hoover					Project Address			FWRH - Refuge Headquarters, Merritt Island, FL 32955	
Phone	321-622-0830									
Email	office@bioprofl.com					Analyzed by/ Date			SR	05/26/26
Lab ID Number	M-JADA-265637-004			M-JADA-265637-005			M-JADA-265637-006			M-JADA-265637-007
Collection Date	05/22/26			05/22/26			05/22/26			05/22/26
Volume	15			15			15			N/A
Location	East Office Wall			Shared Wall			AC Vent			W. Office Wall
RESULT†	**ELEVATED**			**NOT ELEVATED**			**NOT ELEVATED**			**GROWTH LIKELY**
% Slide Analyzed	100			100			100			100
Spore Identification	Raw Count	Spore/m³	% of Total	Raw Count	Spore/m³	% of Total	Raw Count	Spore/m³	% of Total	Spore Level
Aspergillus/ Penicillium	18	1200	95		0	0		0	0	
Chaetomium		0	0		0	0		0	0	
Stachybotrys	1	67	5		0	0		0	0	Medium
Alternaria		0	0		0	0		0	0	
Arthrinium		0	0		0	0		0	0	
Ascospores		0	0		0	0		0	0	
Basidiospores		0	0		0	0		0	0	
Cladosporium		0	0		0	0		0	0	
Cercospora		0	0		0	0		0	0	
Curvularia		0	0		0	0		0	0	
Dreschlera/ Bipolaris		0	0		0	0		0	0	
Epicoccum		0	0		0	0		0	0	
Fusarium		0	0		0	0		0	0	
Ganoderma		0	0		0	0		0	0	
Memnoniella		0	0		0	0		0	0	
Myxomycetes/ Smut		0	0		0	0		0	0	
Nigrospora		0	0		0	0		0	0	
Pithomyces		0	0		0	0		0	0	
Rust		0	0		0	0		0	0	
Spegazzinia		0	0		0	0		0	0	
Torula		0	0		0	0		0	0	
Ulocladium		0	0		0	0		0	0	
Other		0	0		0	0		0	0	
Total Fungi	19	1267	100	0	0	100	0	0	100	N/A
Hyphal Fragment		0	N/A		0	N/A		0	N/A	Low
Background Debris (1-5)*		4			4			3		1

Background Debris is a subjective assessment of the debris level (i.e., house dust) present in the sample, ranked from 1 to 5. A higher number corresponds to a higher level of debris.

*Higher Background Debris may interfere with the analyst's ability to identify spores

1 = 0-5% debris; 2 = 5-25% debris; 3 = 25-75% debris; 4 = 75-90% debris; 5 = 90-100% debris

AIR RESULT KEY†

ELEVATED	The concentration of spores in this sample exceeds the upper HHS threshold, which indicates that an indoor mold source is LIKELY.
POSSIBLE	The concentration of spores in this sample exceeds the lower HHS threshold, which indicates that an indoor mold source is POSSIBLE.
NOT ELEVATED	The concentration of spores in this sample does not exceed the lower HHS threshold, which indicates that an indoor mold source is UNLIKELY.
CONTROL	The indoor samples are compared to the control sample to determine whether there may be an indoor mold source.

SURFACE RESULT KEY†

GROWTH LIKELY	ASSOCIATED WITH IICRC S520 CONDITION 3: ACTIVE MOLD	**SURFACE LEVELS**	
GROWTH POSSIBLE	ASSOCIATED WITH IICRC S520 CONDITION 2: SETTLED SPORES	Rare: 1-9 spores	Low: 10-100 spores
GROWTH UNLIKELY	ASSOCIATED WITH IICRC S520 CONDITION 1: NORMAL ECOLOGY	Medium: 101-1,000 spores	High: >1,000 spores

The Laboratory is not responsible for project sampling. Customer provided information: Project Name, Project Number, Project ID, Project Address, Collection Date, Volume, and Location

Method : ASTM D7391

4795 Enterprise Ave
Naples, Florida 34104
login@daanelabs.com
www.daanelabs.com

REPORT CODE: M-JADA-265637

Company	Bio Pro		Project Name	FWRH
Address	1275 S Patrick Dr H4, Satellite Beach, FL 32937			
Contact	Joy Hoover		Project Address	FWRH - Refuge Headquarters, Merritt Island, FL 32955
Phone	321-622-0830			
Email	office@bioprofl.com		Analyzed by/ Date	SR 05/26/26
Lab ID Number	M-JADA-265637-008	Intentionally Left Blank	Intentionally Left Blank	Intentionally Left Blank
Collection Date	05/22/26			
Volume	N/A			
Location	Attic			
RESULT†	**GROWTH POSSIBLE**			
% Slide Analyzed	100			
Spore Identification	Spore Level			
Aspergillus/ Penicillium				
Chaetomium				
Stachybotrys	Low			
Alternaria				
Arthrinium				
Ascospores				
Basidiospores				
Cladosporium	Low			
Cercospora				
Curvularia				
Dreschlera/ Bipolaris				
Epicoccum				
Fusarium				
Ganoderma				
Memnoniella				
Myxomycetes/ Smut				
Nigrospora				
Pithomyces				
Rust				
Spegazzinia				
Torula				
Ulocladium				
Other				
Total Fungi	**N/A**			
Hyphal Fragment	Low			
Background Debris (1-5)*	1			

Background Debris is a subjective assessment of the debris level (i.e., house dust) present in the sample, ranked from 1 to 5. A higher number corresponds to a higher level of debris.

*Higher Background Debris may interfere with the analyst's ability to identify spores

1 = 0-5% debris; 2 = 5-25% debris; 3 = 25-75% debris; 4 = 75-90% debris; 5 = 90-100% debris

AIR RESULT KEY†

ELEVATED	The concentration of spores in this sample exceeds the upper HHS threshold, which indicates that an indoor mold source is LIKELY.
POSSIBLE	The concentration of spores in this sample exceeds the lower HHS threshold, which indicates that an indoor mold source is POSSIBLE.
NOT ELEVATED	The concentration of spores in this sample does not exceed the lower HHS threshold, which indicates that an indoor mold source is UNLIKELY.
CONTROL	The indoor samples are compared to the control sample to determine whether there may be an indoor mold source.

SURFACE RESULT KEY†

GROWTH LIKELY	ASSOCIATED WITH IICRC S520 CONDITION 3: ACTIVE MOLD	**SURFACE LEVELS**	
GROWTH POSSIBLE	ASSOCIATED WITH IICRC S520 CONDITION 2: SETTLED SPORES	Rare: 1-9 spores	Low: 10-100 spores
GROWTH UNLIKELY	ASSOCIATED WITH IICRC S520 CONDITION 1: NORMAL ECOLOGY	Medium: 101-1,000 spores	High: >1,000 spores

The Laboratory is not responsible for project sampling. Customer provided information: Project Name, Project Number, Project ID, Project Address, Collection Date, Volume, and Location

Method : ASTM D7391

BIO PRO
Mold & IAQ Testing

1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

Lab Samples & Results

• East Office N. Wall: Air Sample – Elevated (Aspergillus/Penicillium, Stachybotrys)

• Attic: Swab Sample – Growth Possible (Stachybotrys, Cladosporium)

• W Office Wall: Swab Sample – Elevated (Stachybotrys)

• AC Vent: Air Sample – Not Elevated

• Shared Wall: Air Sample – Not Elevated

• West Wall: Air Sample – Not Elevated

• East Wall: Air Sample – Not Elevated

• Conference Table: Air Sample – Not Elevated

Conclusions

The inspection and laboratory testing indicate that mold growth is present in several localized areas, particularly in the attic and on the north and west walls of the east and west offices. While visible **mold was confirmed by surface and wall cavity samples, air quality testing in occupied spaces and HVAC vents did not show elevated airborne mold levels, suggesting that the contamination is not currently widespread in the indoor air**. High indoor humidity and evidence of water intrusion are consistent with the observed mold growth and are likely contributing factors.

For the client, this means that **targeted remediation is necessary to address the identified mold-impacted materials and correct the underlying moisture issues.** The presence of mold is consistent with previous water damage and ongoing high humidity, which should be addressed to prevent future problems. By removing affected materials, thoroughly cleaning impacted areas, and controlling humidity and water intrusion, the risk of mold recurrence can be significantly reduced. Clearance testing is recommended after remediation to ensure a healthy environment before employees return to the affected spaces.

BIO PRO
Mold & IAQ Testing

1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

Remediation Recommendations

1. MOISTURE AND HUMIDITY CONTROL

- Investigate and repair all sources of water intrusion, especially at the north wall and window casings
- Reduce indoor humidity to 50-55% using dehumidifiers or HVAC adjustments before and during remediation

2. CONTAINMENT AND SAFETY

- Establish proper containment barriers in the east and west offices to prevent cross-contamination
- Use negative air machines with HEPA filtration during removal and cleaning activities

3. REMOVAL OF CONTAMINATED MATERIALS

- Remove and discard **the entire north wall in both the east and west offices, as well as the west wall in the west office up to 2 feet high**
- Remove and discard the affected attic roof decking sections with visible mold growth
- Remove and replace the contaminated flex duct branch in the conference room

4. CLEANING AND SANITIZATION

- Thoroughly HEPA vacuum and apply antimicrobial treatments to all exposed surfaces behind removed materials
- Clean all non-porous and semi-porous surfaces in affected areas
- Clean or replace AC registers with visible mold growth

5. POST-REMEDIATION VERIFICATION

- Conduct clearance testing to confirm that mold levels are within acceptable limits before reoccupying the remediated areas
- Monitor humidity and inspect for any signs of recurring moisture or mold

1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

Remediation Recommendations

Red Arrow - Location of Elevated Mold Sample.

Green Arrow - Location of Acceptable Mold Sample.

Red Walls - Remove & Discard drywall and insulation from floor to ceiling.

Orange Walls - Remove & Discard drywall and insulation 2ft. From the floor and inspect for signs of continued damage (staining, growth, etc.). Continue removing materials 24” beyond the last signs of visible damage.



BIO PRO
Mold & IAQ Testing

1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

Recommended Remediation Standards

Containment	Place a 6-mil polyethylene containment barrier at the entrance of the work area with an entry/egress cover flap on the outside. Shut down or isolate HVAC system within containment area by sealing any return air vents, registers, or exhaust systems. Maintain containment area under negative pressure relative to the surrounding area outside containment using HEPA-filtered Air Filtration Devices (AFD's). General industrial hygiene practices recommend a minimum of four (4) air changes per hour for containment ventilation and dilution. AFD's must be shut off 1-24hr prior to PRV testing according to IICRC standards.
Contents	Remove and discard anything that is porous and sterilize anything that is non-porous (this is at the remediator's discretion).
Contaminated Materials	All contaminated materials should be removed & discarded in well-sealed plastic bags while still inside containment. Materials should be removed 24in. beyond last visible damage. During the remediation process, additional areas of water damage or mold growth may be discovered & should be addressed. Addendums to this Protocol will be written if deemed necessary.
Detail Cleaning	HEPA Vacuum any remaining building materials

1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

	within the workspace. Dispose of vacuum contents in well-sealed plastic bags while still inside the containment. Sand, Wire Brush, and Wet Wipe-Down all remaining exposed building materials (studs, framing, slab, etc.) and sanitize the entire workspace with approved fungicidal applicant. Severely damaged framework should be removed and replaced (visible wood rot or deterioration beyond the point of saving with cleaning).
Personal Protective Equipment	At a minimum, N95 Respirators, Eye Protection, and Gloves must be worn at all times during remediation. For more information regarding PPE, please refer to the IICRC-S520.
Clearance Testing	All AFD's, air movers, or other equipment should be turned off 1-24hr prior to PRV testing, according to IICRC standards. PRV testing must be conducted while containment is still in place.

BIO PRO
Mold & IAQ Testing

1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

AC System Recommendations

- At a minimum, the interior of the air handler should be professionally cleaned (including the coils, fan, and insulation) using HEPA filtered vacuum cleaners and wet wiping with an appropriate cleaner. All air handler interior insulation should be removed and replaced.

- Thorough cleaning should include the removal of the fan component to fully access the interior of the air handler. The exterior of the air handler and corresponding surfaces should be HEPA filter vacuumed and wet wiped with an appropriate cleaner.

Preventing Mold in HVAC Systems

Mold growth inside air conditioning (AC) systems is primarily due to two factors:

- Excess moisture
- Accumulation of dust or debris.

Mold thrives in damp conditions, and when combined with dust and debris that can serve as a food source, these spores can quickly multiply and lead to mold contamination. The enclosed nature of AC systems, along with the fact that they often create condensation, make them an ideal breeding ground for mold. Below are some tips to help prevent future mold growth after your AC system + duct cleaning or replacement:

Regularly Change & Upgrade Filters	Your AC filter is the first line of defense against dust and debris. Regularly changing the filter (every 1-3 months, depending on usage and filter type) can prevent this debris from accumulating and supplying food for mold. Consider upgrading to a filter with a high MERV rating for more effective filtration of small particles, including mold spores.
Monitor & Control Humidity Levels	Mold thrives in high humidity. Using a dehumidifier, or an AC system with built-in humidity control, can help maintain a relative humidity level of 30-50%, which is unfavorable for mold growth. Regularly check the condensation drain pan and drain line to make sure they are draining properly and not contributing to excess moisture in the system.

1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

Mold Glossary

Ascospores:

A large category of spores (produced in a sac-like structure called an ascus) that are found everywhere in nature and include more than 3000 genera. Most Ascospores of health or IAQ importance are identified separately by their genus (e.g. Chaetomium) when possible on a IAQ report, and the Ascospore category is used primarily on these reports for a large group of less important spore types often found in quantity on outdoor air samples. On tape samples, Ascospore is sometimes also used as a general morphological identification (i.e. the ascus or sac structure is present) for certain samples in those cases when the spores do not appear to represent any of the IAQ significant genera.

Aspergillus/Penicillium:

Allergen, contaminant, and opportunistic pathogen. It comprises approximately 200 species and can appear almost any color. Aspergillus/Penicillium is known to be the most common fungal species in water damaged properties. Health effects vary by species, but many species are reported to be allergenic. Some species produce toxins that might have significant health effects in humans. Immunosuppression is the major factor predisposing to development of opportunistic infections. Construction in hospital environments constitutes a major risk for development of aspergillosis particularly in immunocompromised patients. Aspergillus versicolor can be found mostly in temperate areas in air, house dust, foods, soils, hay, cotton, and dairy products. Its presence in indoor air often indicates signs of moisture problems in buildings, as it is readily found in water damaged building materials. This species produces the mycotoxin sterigmatocystin, which is reported to be carcinogenic to the liver and kidney, and it can cause such symptoms as diarrhea and upset stomach. It also produces the volatile organic compound (VOC) geosmin, this compound causes irritation of the mucus membranes of humans and pets; also causing the characteristic musty, earthy odor often connected with moldy houses. Known to cause Type I and Type III fungal hypersensitivity.

Basidiospores:

Basidiospores are known as “higher fungi” or true mushrooms and are a general category of spores that have been released from the basidium of a fungus. This group of spores is ubiquitous in nature and is mainly found in gardens, forests, and woodlands. Basidiospores are moisture driven as their spores disseminate during rain or in times of high humidity. The presence of these spores at levels greater than those of the outdoor environment may be an indicator of fungal contamination and

BIO PRO
Mold & IAQ Testing

1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

water damage inside a building. Other members of the Phylum Basidiomycota include shelf fungi, rusts, and smuts. Known to cause Type I and Type III fungal hypersensitivity.

Bipolaris/Drechslera:

Contaminant and opportunistic pathogen found on soils, grasses, plants, and decaying food. Known to cause Type I fungal hypersensitivity and the most common agent for allergic fungal sinusitis. Various but uncommon infections of the eye, nose, lungs, and skin have been reported.

Chaetomium:

Contaminant that is rarely involved in systemic and cutaneous disease and sometimes reported to produce harmful allergic toxins. Primary IAQ importance is that it will grow in the same conditions as Stachybotrys (wet cellulose) and amplified amounts in indoor air could be a warning that conditions do exist for Stachybotrys growth. Many times on damp sheetrock paper, colonies of Chaetomium and Stachybotrys will be growing on top of one another or side by side. Known to cause Type I fungal hypersensitivity.

Cladosporium:

Cladosporium is commonly observed indoors with excess moisture or humidity. It tends to grow in areas like bathrooms, kitchens, and attics that are prone to dampness and water damage, but is also frequently observed in air conditioning systems that have moisture or condensation issues. If the AC system also contains excess dust or debris, Cladosporium can grow and spread quickly. In some individuals, exposure to Cladosporium can lead to symptoms consistent with fungal hypersensitivity Types I, II, and III.

Curvularia:

Contaminant and opportunistic pathogen found in air, soil and textiles. Reported to be allergenic. Rare infections of corneas, nails, and sinuses, primarily in immunocompromised individuals. Known to cause Type I fungal hypersensitivity.

1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

Epicoccum:

Contaminant and opportunistic pathogen, found in soil, air, water and rotting vegetation and can be commonly found in outdoor air. It is a common allergen, and rarely can it cause an infection in the skin. Known to cause Type I Fungal Hypersensitivity.

Stachybotrys:

Contaminant, found indoors primarily on wet cellulose containing materials (GWB, textiles, etc.). Stachybotrys is sometimes difficult to detect indoors because many times it will grow unseen on the back of walls or in the wall cavity with little disturbance that would cause it to be detected by routine air sampling. This is potentially also when it is of most health concern: when it covers entire wall areas and constantly produces toxins undetected. It has the ability to cause type I hypersensitivity. It is a documented mycotoxin producer. Stachybotrys may play a role in the development of sick building syndrome. The presence of this fungus can be significant due to its ability to produce mycotoxins. Known to cause Type I fungal hypersensitivity.

Myxomycete/Rust/Smut:

General category for commonly found genera usually associated with living and decaying plants as well as decaying wood. Sometimes can be found indoors. Some allergenic properties reported, but generally pose no health concerns to humans or animals. Smuts are ubiquitous parasitic plant pathogens, which require a living host to complete their life cycle and are not usually found growing indoors. Smuts are most often found on corn, grass, weeds, flowering plants and other fungi; they are usually disseminated by wind. They are called smuts because they form black powdery spore masses that resemble soot or smut. Myxomycetes are ubiquitous outdoor molds (often considered a slime mold), and are commonly found in forested areas where bark, soil, dung, and leaf litter are present.

References:

- Bioaerosols: Assessment and Control. Janet Macher, Ed., American Conference of Governmental Industrial Hygienists, Cincinnati, OH 1999

- Exposure Guidelines for Residential Indoor Air Quality. Environmental Health Directorate, Health Protection Branch, Health Canada, Ottawa, Ontario, 1989

- Fungal Contamination in Public Buildings: Health Effects and Investigation Methods. Health Canada, Ottawa, Ontario, 2004

- IICRC: S500 Standard and Reference Guide for Professional Water Damage Restoration. 3rd Edition, Institute of Inspection, Cleaning, and Restoration Certification, Vancouver, WA, 2006

BIO PRO
Mold & IAQ Testing

1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

Mold Sampling Methods

Bio Pro utilizes spore trap impacted air samples and swab/tape lift surface samples to collect and analyze for genus identification. Each method of sampling presents its own advantages and disadvantages, and it is our responsibility to determine which method is best suited for your specific job.

Sample Type	Description	About the Lab Results
Air Samples	Air samples are typically evaluated by means of fungal type identified and by comparing indoor to outdoor concentrations, complaint to non-complaint areas, or areas of concern to areas of non-concern. In general, the levels and types of fungi in the indoor air (in non-problem buildings) should be similar to, or lower than, those found in the outdoor air. Higher levels of spores found inside may indicate that moisture sources and resultant fungal growth are present. Spore count results are influenced by geographic location, seasonal variation, and other outside conditions.	*Use the following guidelines when interpreting Lab Results:* 1. The composition and diversity of fungi in indoor, non-problem buildings should be similar to that of the outdoor air. 2. Compare spore count concentrations indoors and outdoors. Elevated indoor concentrations may indicate indoor fungal growth. Be aware that this is not always consistent and additional sampling may be needed. 3. Certain fungi are very good indicators of water damage. The presence of these spores, even in small quantities, indicates the presence of water damage. These indicator fungi include, but are not limited to, Aspergillus/Penicillium, Stachybotrys (including Memnoniella), and Chaetomium.
Swab/Tape Lift Samples	The presence of common spores in the “rare” or “low” categories in surface samples suggests only background deposition and not growth. Categories greater than this or the presence of fungal vegetative and/or reproductive fragments (e.g. hyphae) suggests fungal colonization, growth, and/or accumulation at or near the sampled location.	The presence of water damage associated fungi is also an indicator that indoor growth may be occurring. These indicator fungi include, but are not limited to, Chaetomium, Fusarium, Stachybotrys (including Memnoniella), and Aspergillus.

BIO PRO
Mold & IAQ Testing

1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

Understanding Wall Cavity Sampling

As part of your mold inspection, we may have taken air samples from inside your walls. These samples are used to detect mold that may not be visible during a standard inspection and provide valuable insight into the condition within the wall cavity.

However, wall cavities often contain dust, insulation, and other debris that can mix with the air sample. When this happens, it can interfere with the clarity of the results. If a sample is heavily contaminated with debris, it may be difficult or even impossible to accurately identify and quantify mold spores. In some cases, we can still determine the presence of certain mold types, but not the concentration.

To minimize this interference, we use handheld vacuums to clean out sampling areas as best we can before collecting air. Even with these efforts, some debris may remain and affect the sample. When a sample result is inconclusive, we may recommend retesting, using other types of inspection methods, or creating small access points to visually inspect behind the drywall.

We are committed to delivering accurate and clear mold assessments. While environmental conditions inside walls can present some challenges, we will always guide you through the best next steps to help you understand the true condition of your home.

Disclaimer

Fungal spores are found everywhere. Whether or not symptoms develop in people exposed to fungi depends on the nature of the fungal material (e.g. allergenic, toxic, or infectious), the exposure level, and the susceptibility of exposed persons. Susceptibility varies with the genetic predisposition, age, pre-existing medical conditions (diabetes, cancer, immunosuppression, or chronic lung conditions), use of immunosuppressive drugs, and concurrent exposures. These reasons make it difficult to identify dose/ response relationships that are required to establish “safe” or “unsafe” levels (i.e. Permissible Exposure Limits). There are no standard Permissible Exposure Limits or Threshold Level Values for fungal contamination.

It is generally accepted in the industry that indoor fungal growth is undesirable and inappropriate, necessitating removal or other appropriate remediation actions. The New York City Guidelines, EPA guidelines, and IICRC guidelines define the conditions warranting mold remediation. Always remember that water/moisture is the key. Preventing water damage or water condensation will aid the prevention of mold growth.

1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

This report is not intended to provide medical advice or advice concerning the relative safety of an occupied space. Always consult an occupational or environmental health physician who has experience addressing indoor air contaminants if you have any questions.

Moisture Readings

At Bio Pro, we use an Extech M0290 dual function moisture meter with infrared thermometer and psychrometric measurement capabilities to conduct all moisture readings. Designed for both invasive and non-invasive testing, this device allows us to measure the amount of moisture in all types of building materials. The industry-wide standard is that building materials in excess of 15% moisture content could indicate a water/moisture problem. Often times we are unable to gather moisture readings from non-visible areas and therefore can only draw limited conclusions about the moisture of any area tested. It is our responsibility to combine the moisture readings with our industry experience to identify the full scope of the water damage and recommend the appropriate remediation protocol.

1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

Inspection Photos







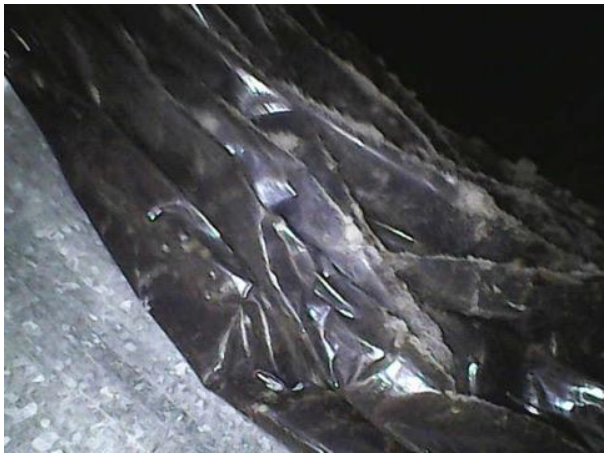






1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

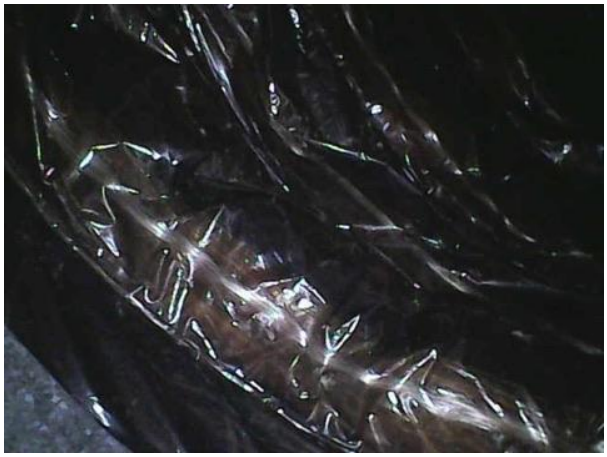







1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830



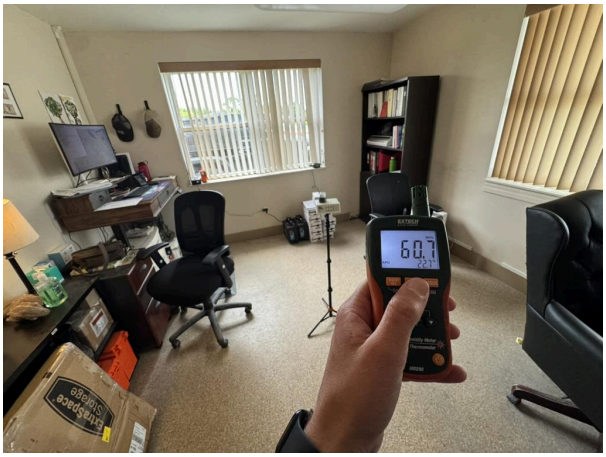





1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830

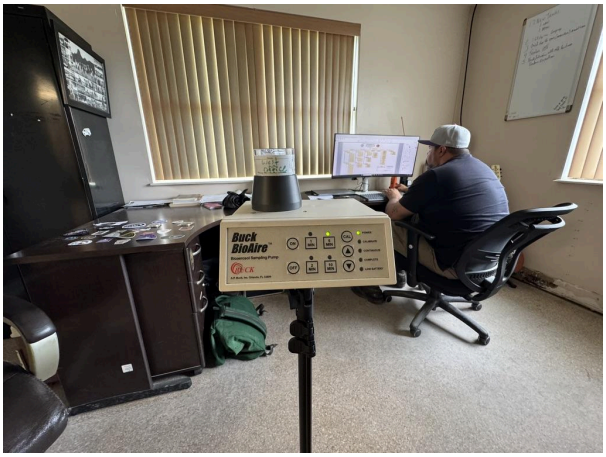




1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830













1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830









1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830






1275 S. Patrick Dr. Ste H4
Satellite Beach, FL 32937
(321) 622-0830





